

 Project 25: Context-Aware and Efficient Data Valuation for Medical Machine Learning

Alpha Design Presentation

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BMEG 457 Biomedical Engineering Design Project

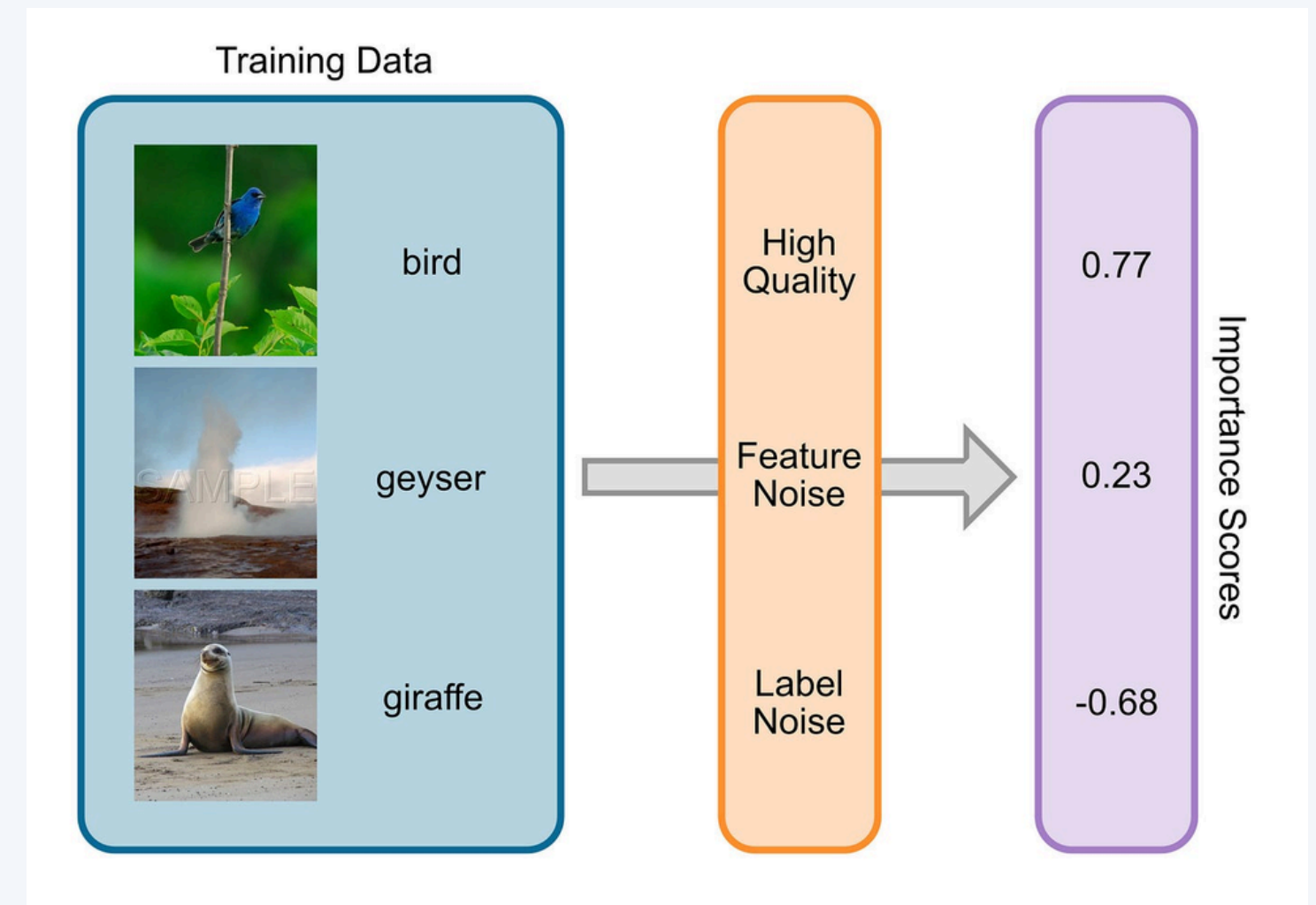
01 Orientation to the Problem

Project Brief

Data Valuation (DV): Quantifies the specific impact of an individual data point on a model's accuracy.

Shapley Values: Calculates a data point's "fair share" of the performance based on its average marginal contribution across all data subsets.

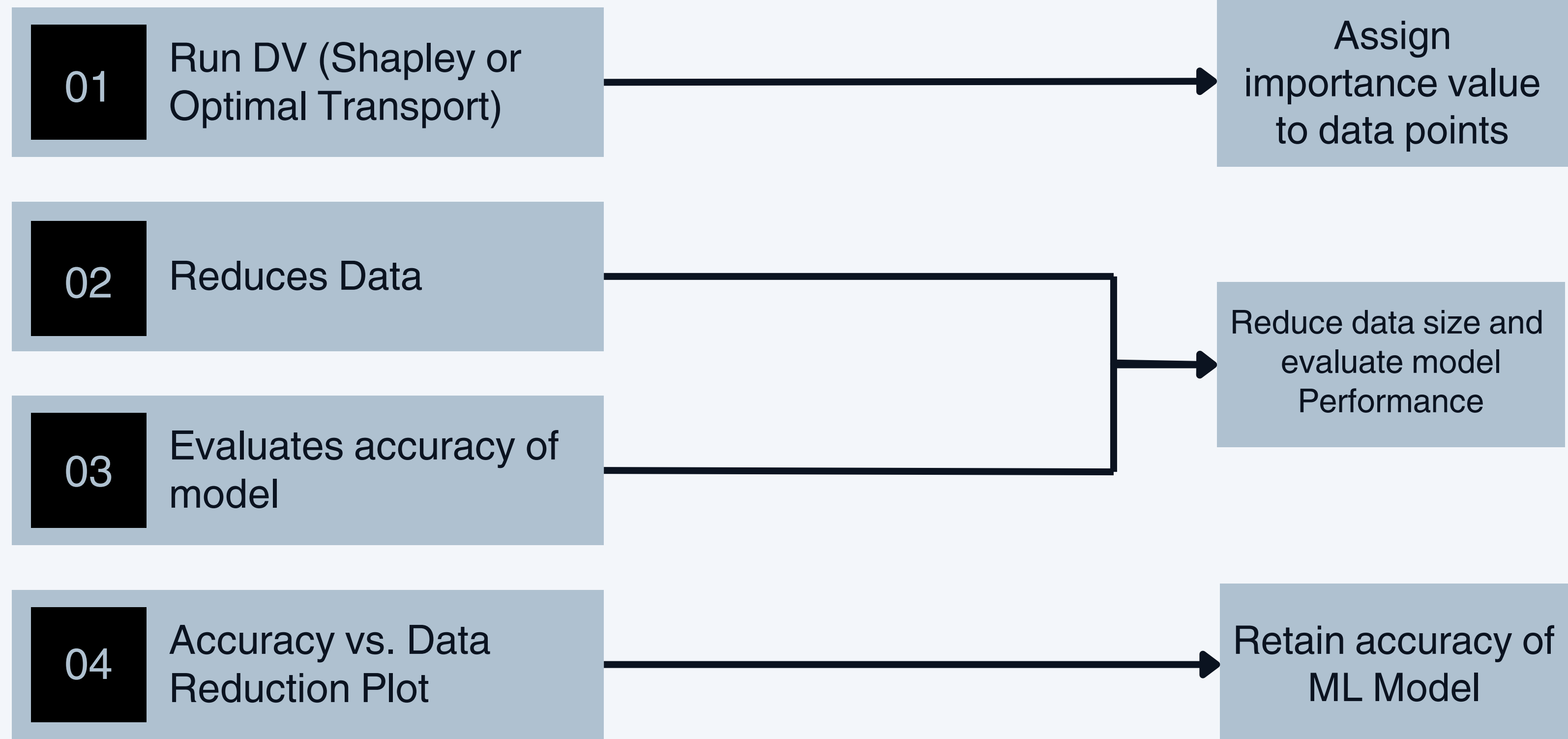
Optimal Transport (OT): Measures the "distance" or effort required to shift a model's learned data distribution to match the true ground-truth distribution.



<https://towardsdatascience.com/data-valuation-a-concise-overview-72e8cf12c755/>

Critical Functions

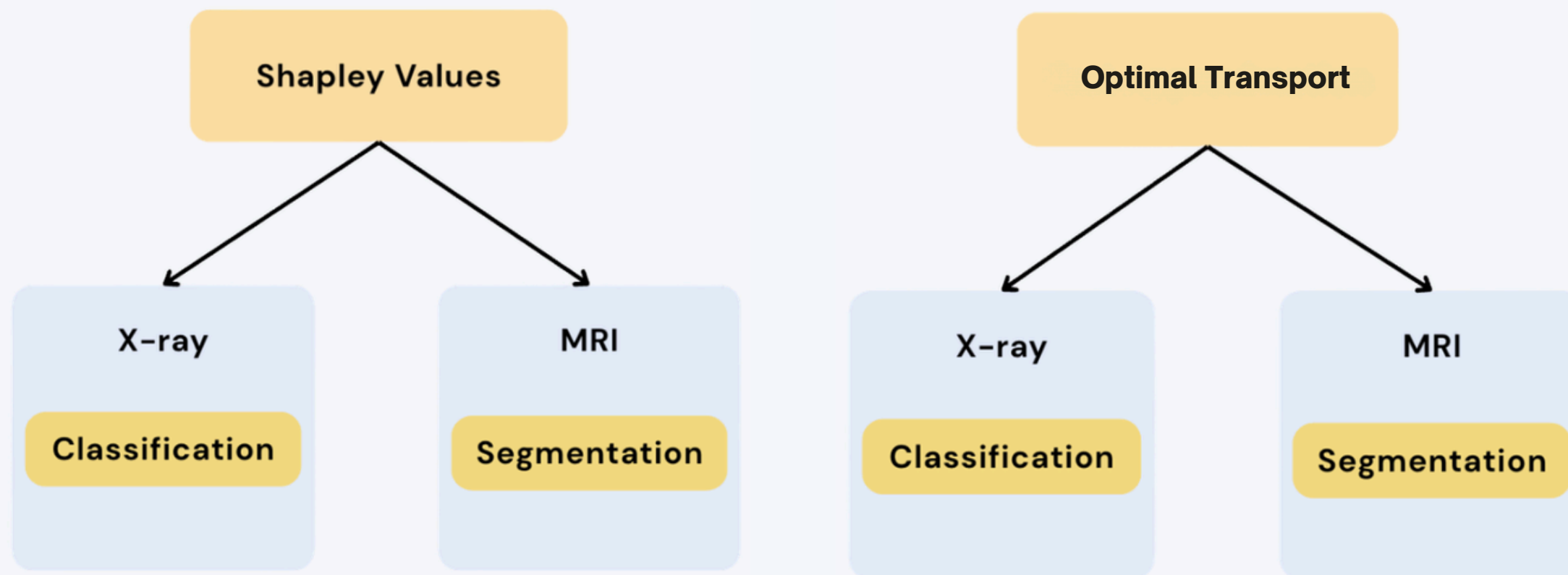
Specifications



02 Build Process

Building Pipelines

Comparative Data Valuation Across Modalities and Tasks



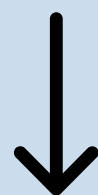
For comparable results of different DV methods:

the same modality and task will use the same dataset and same ML model to build.

*Note: Results across different modality is incomparable

General Work Flow of Each Pipeline

01 Train ML model with 100% of data



Output:
Full model performance
(accuracy / dice)

02 Assign Importance Value to Data Points with DV methods



Output:
Ordered list of images
with importance value

03 Retrain ML model with top valued X% of data and evaluate on test set



Output:
Comparable performance
of model with X% of top
valued data used

03 Alpha Design

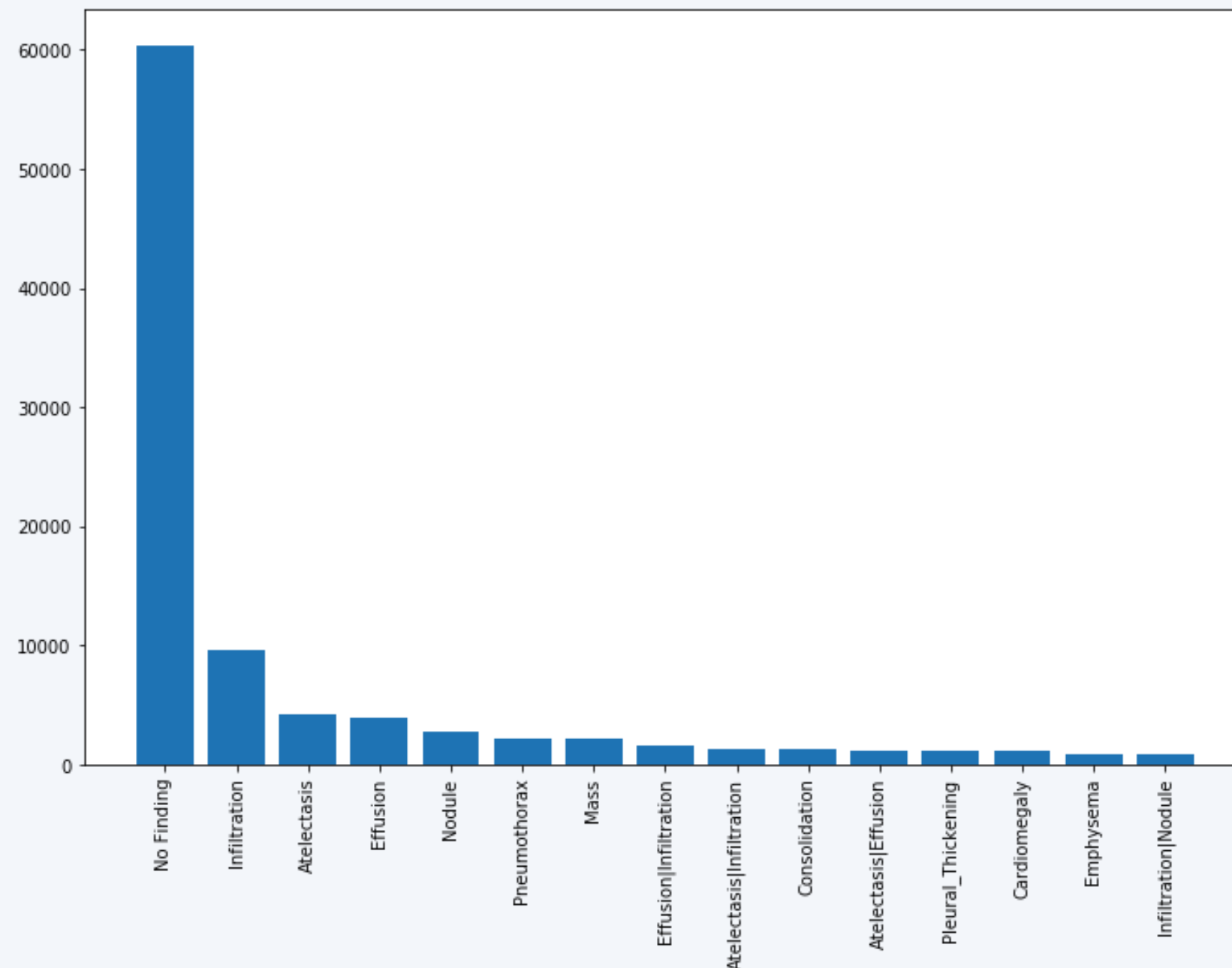
Results

X-ray Classification

Shapley + Optimal Transport

NIH Chest X-rays Data Set

112,120 Images | 30,805 Patients | 14 Disease Labels



Label Creation

Labels extracted from radiology reports
90% estimated label accuracy
Weakly-supervised dataset

Patient-Safe Split

80% Train
10% Validation
10% Test

No patient overlap

DenseNet 121 model Training

DenseNet model summary

```

=====
Layer (type (var_name))                Output Shape                Param #
=====
DenseNet (DenseNet)                    [1, 1]                      --
├─Sequential (features)                 [1, 1024, 7, 7]            --
│   └─Conv2d (conv0)                   [1, 64, 112, 112]         9,408
│   └─BatchNorm2d (norm0)              [1, 64, 112, 112]         128
│   └─ReLU (relu0)                    [1, 64, 112, 112]         --
│   └─MaxPool2d (pool0)                [1, 64, 56, 56]           --
│   └─_DenseBlock (denseblock1)        [1, 256, 56, 56]          335,040
│   └─_Transition (transition1)        [1, 128, 28, 28]          33,280
│   └─_DenseBlock (denseblock2)        [1, 512, 28, 28]          919,680
│   └─_Transition (transition2)        [1, 256, 14, 14]          132,096
│   └─_DenseBlock (denseblock3)        [1, 1024, 14, 14]         2,837,760
│   └─_Transition (transition3)        [1, 512, 7, 7]            526,336
│   └─_DenseBlock (denseblock4)        [1, 1024, 7, 7]           2,158,080
│   └─BatchNorm2d (norm5)              [1, 1024, 7, 7]           2,048
├─Sequential (classifier)               [1, 1]                      --
│   └─Dropout (0)                      [1, 1024]                  --
│   └─Linear (1)                       [1, 1]                     1,025
=====
Total params: 6,954,881
Trainable params: 6,954,881
Non-trainable params: 0
Total mult-adds (Units.GIGABYTES): 2.83
=====

```

Hyperparameters

--epochs: 50

--batch_size: 32

--early_stop_patience: 5

Phase 1 (head training):

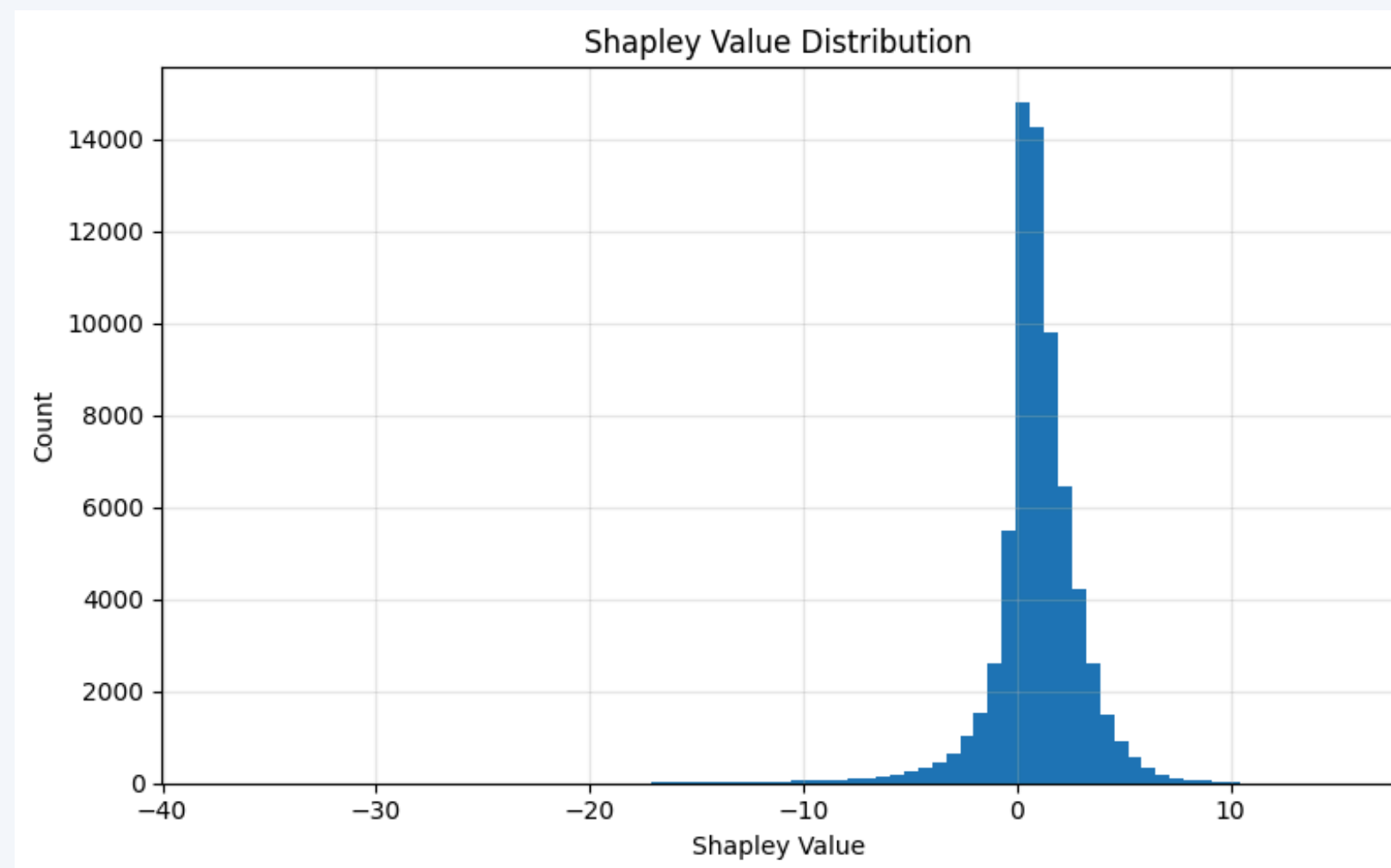
Learning rate: 1e-3

Phase 2 (fine-tuning last block):

Learning rate: 1e-4

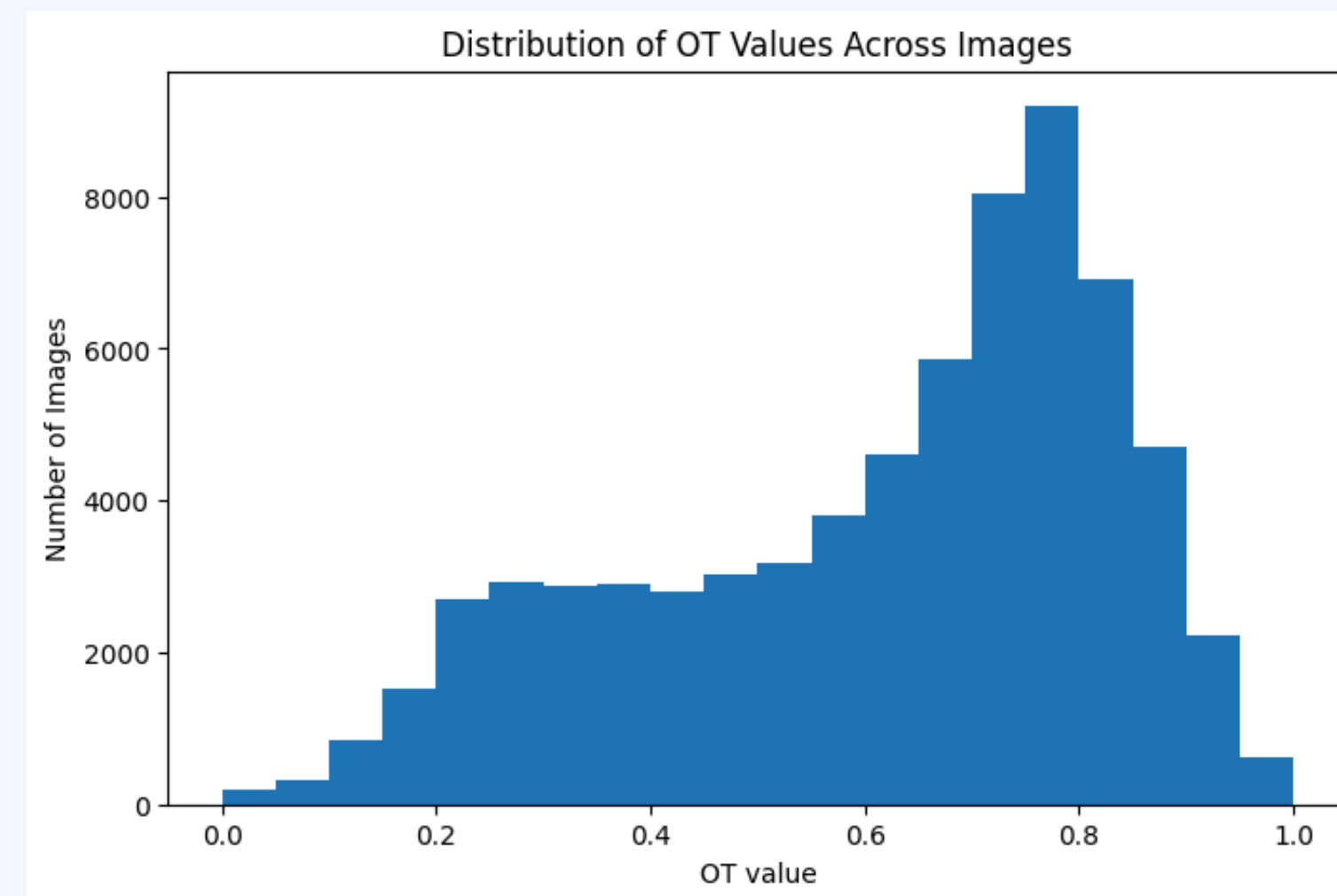
Distribution of Values

Shapley Values



Shapley Values:
Min: -37.41, Max: 15.03
Mean: 0.91, Std: 2.18

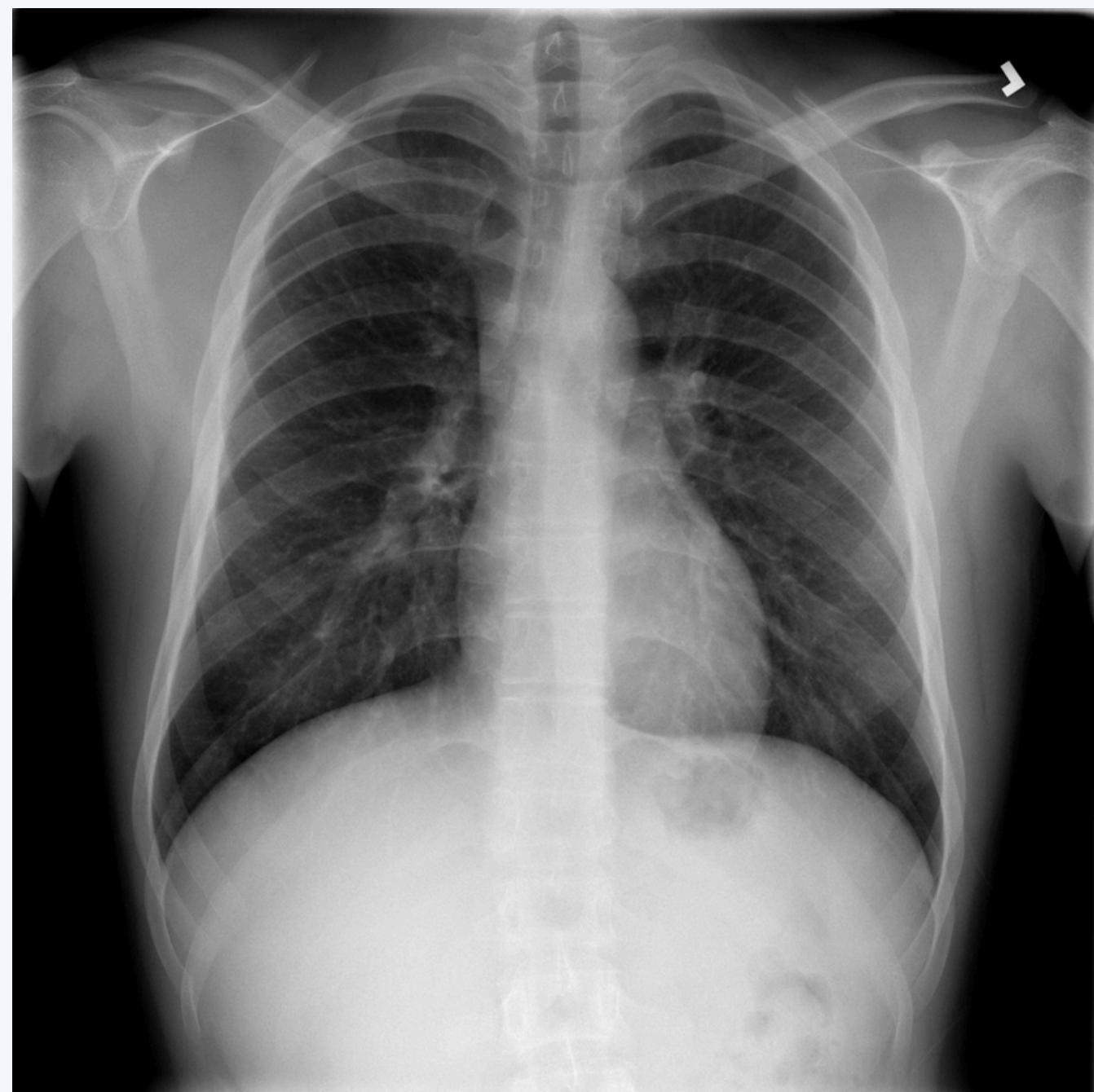
Optimal Transport



Optimal Transport:
min: 0, max: 1
mean: 0.615, std: 0.217

Shapley value: 15

Label: No finding

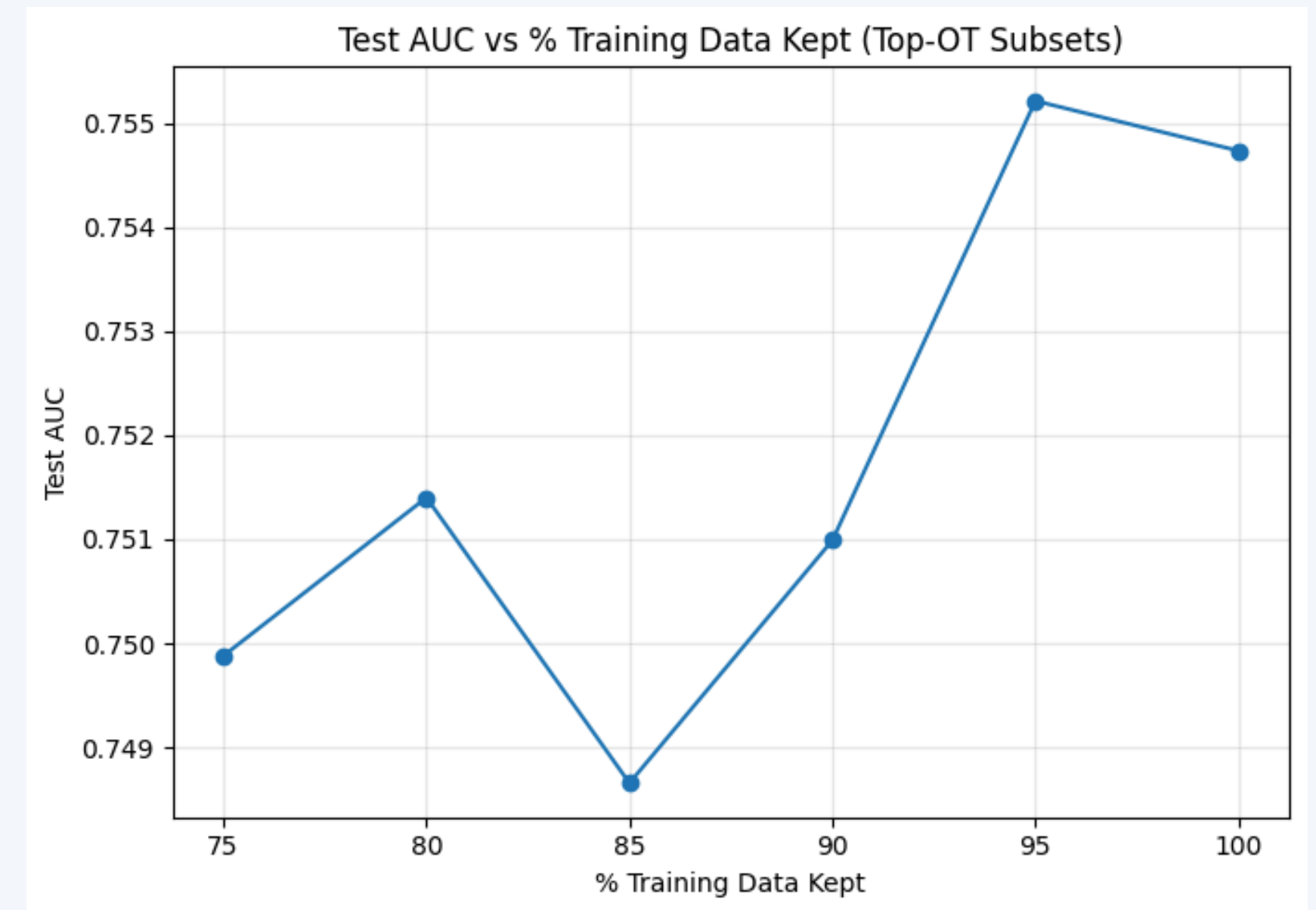
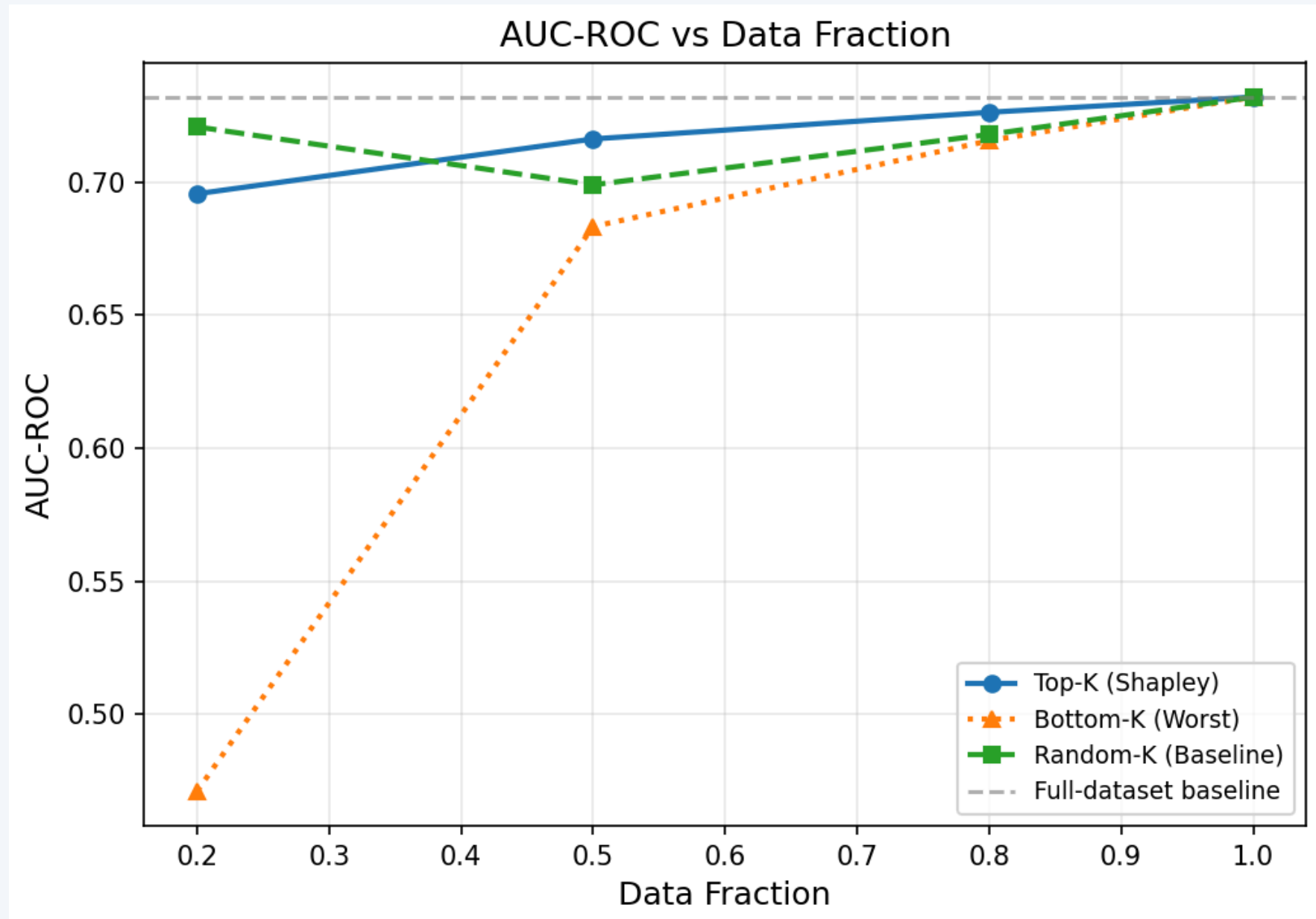


OT value: 0.3

Label: No finding

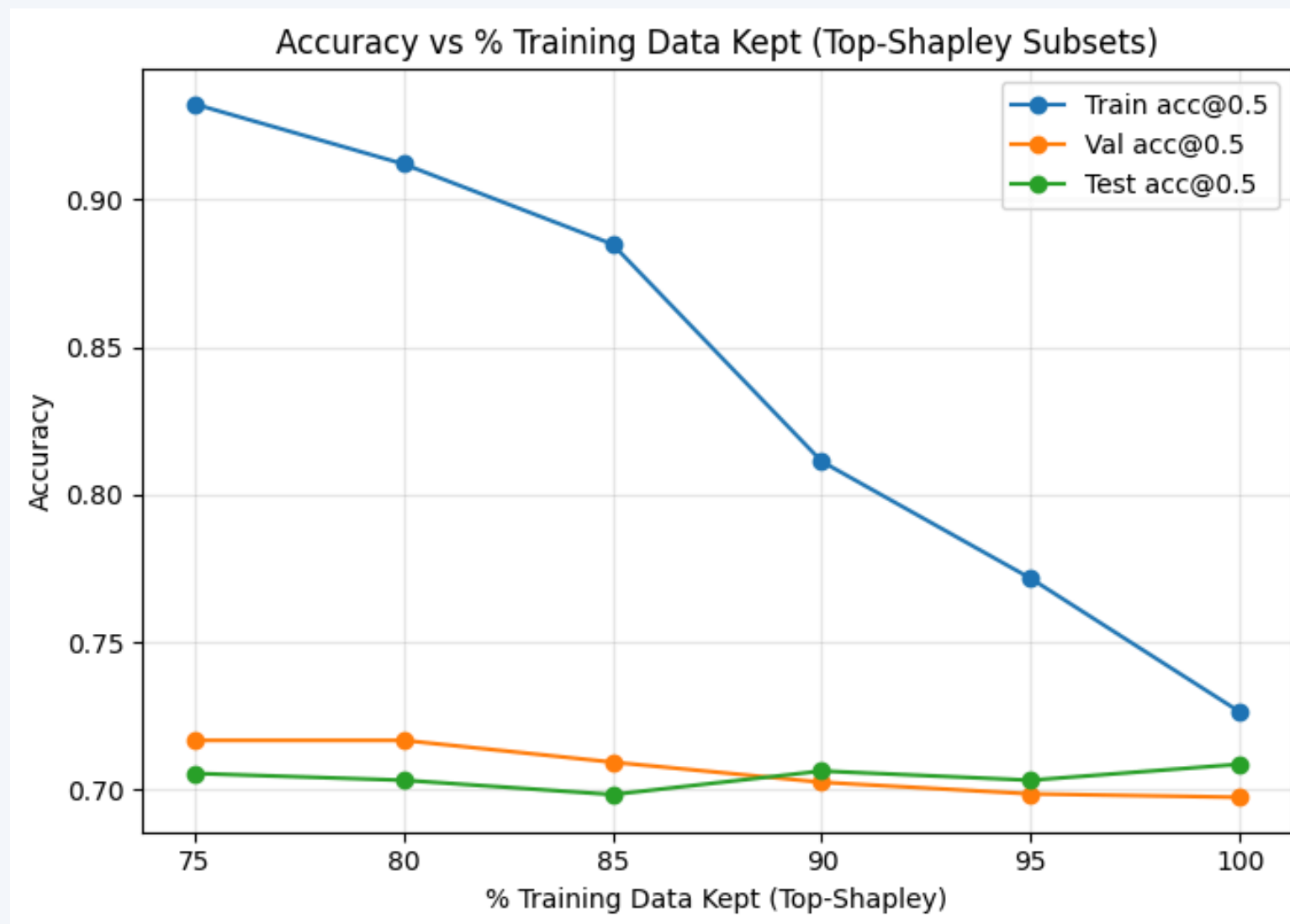


Test AUC vs % Training Data Kept

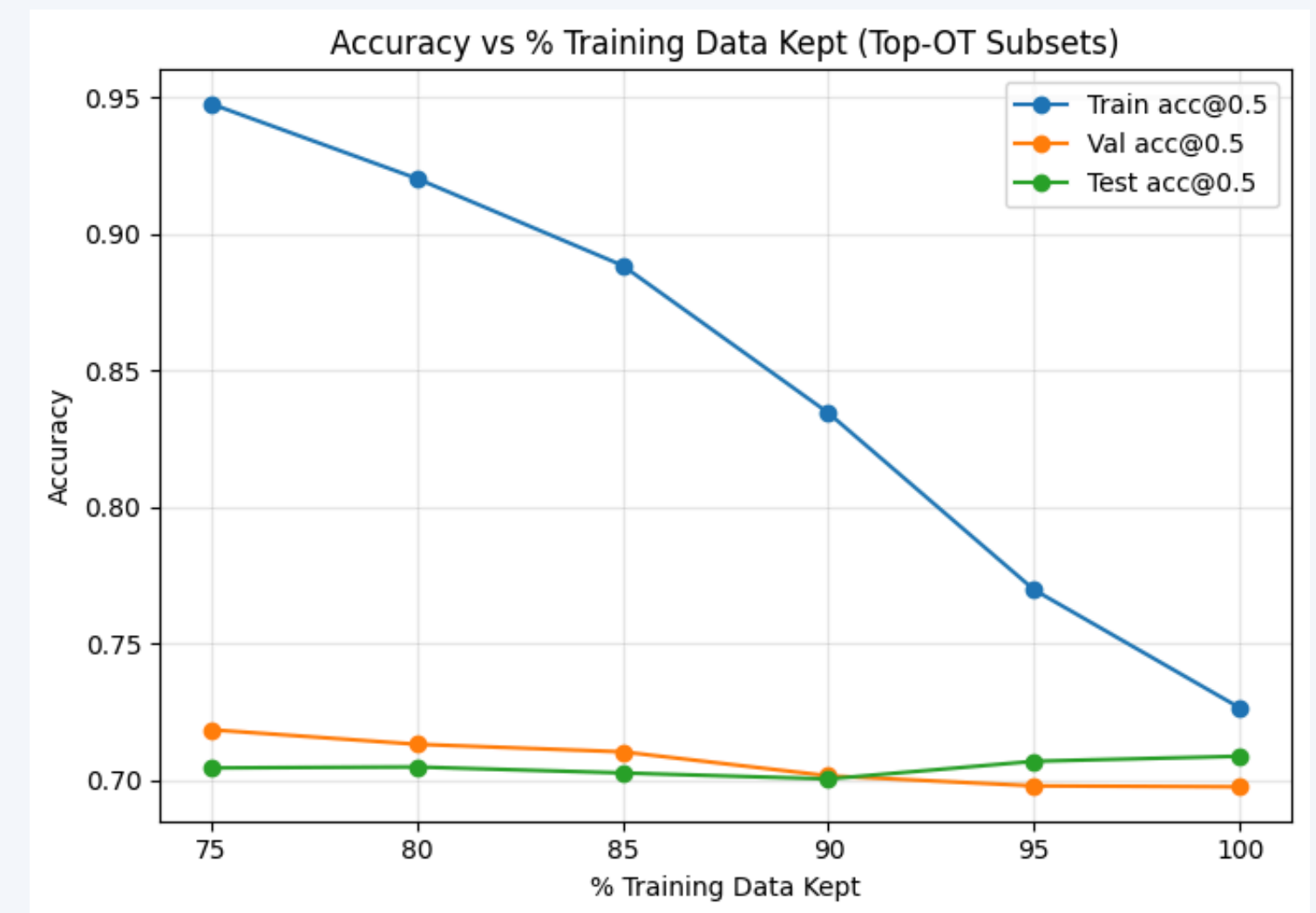


Accuracy vs % Training Data Kept

Shapley Values



Optimal Transport



MRI Segmentation

Shapley + Optimal Transport

Data Set: BraTS 2023 Adult Glioma

- NIfTI files (.nii.gz) 4 modalities
- 4 Labels (multiclass → Binary):
 - 3 GD-enhancing tumour (ET)
 - 2 Peritumoral edematous/invaded tissue (ED)
 - 1 Necrotic tumour core (NCR)
 - 0 Everything else
- Annotated by neuro-radiologists
- Preprocessing applied:
 - co-registered to the same anatomical template
 - Interpolated to the same resolution (1 mm³)
 - Skull-stripped.

Total patients: 1251

- 1 Slice per patient

Split data set:

- 75% Train: 900
- 15% Val: 225
- 10% test: 126

Smaller sample size

expensive capture

03

ML Model - Unet

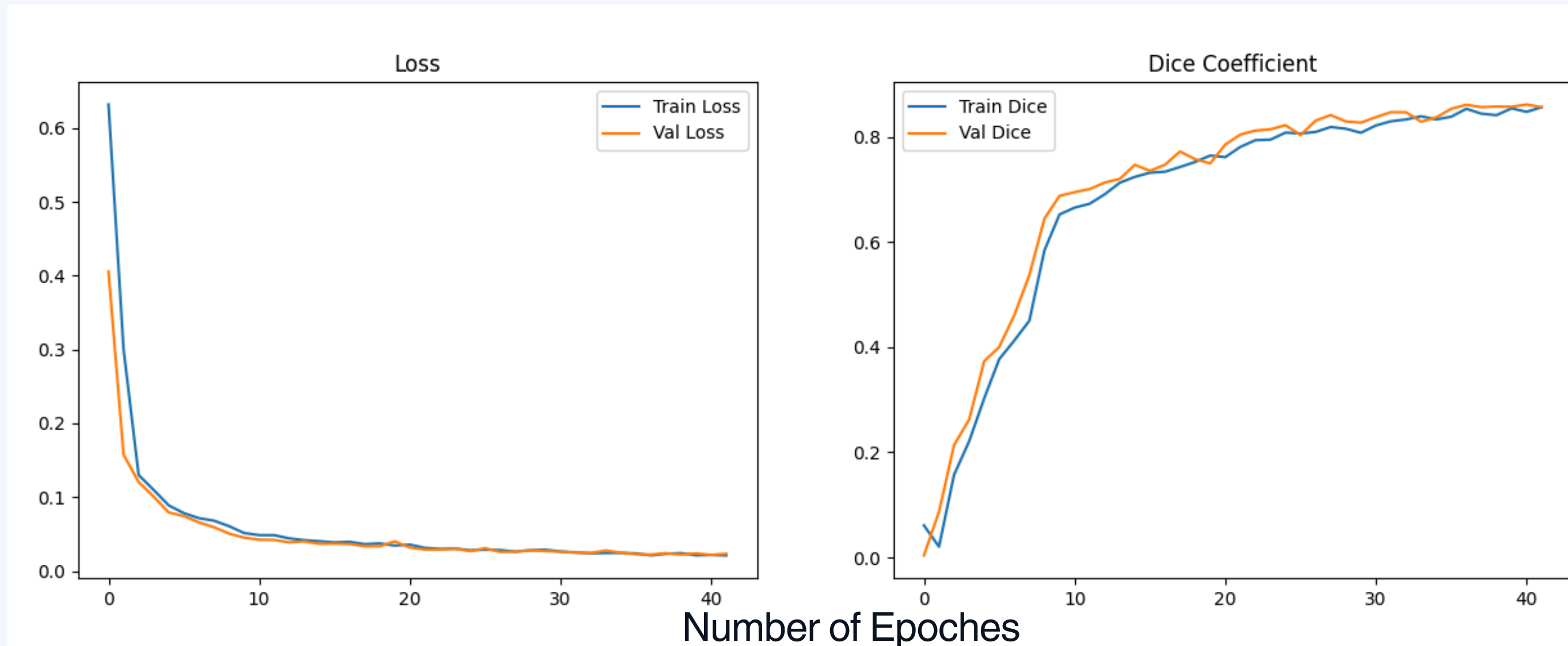
UNet for MRI:

- Based on an established 2023 BRATS Competition pipeline on Kaggle
- High performance on smaller data sets
- Hyperparameter autotuning

DICE Loss Function:

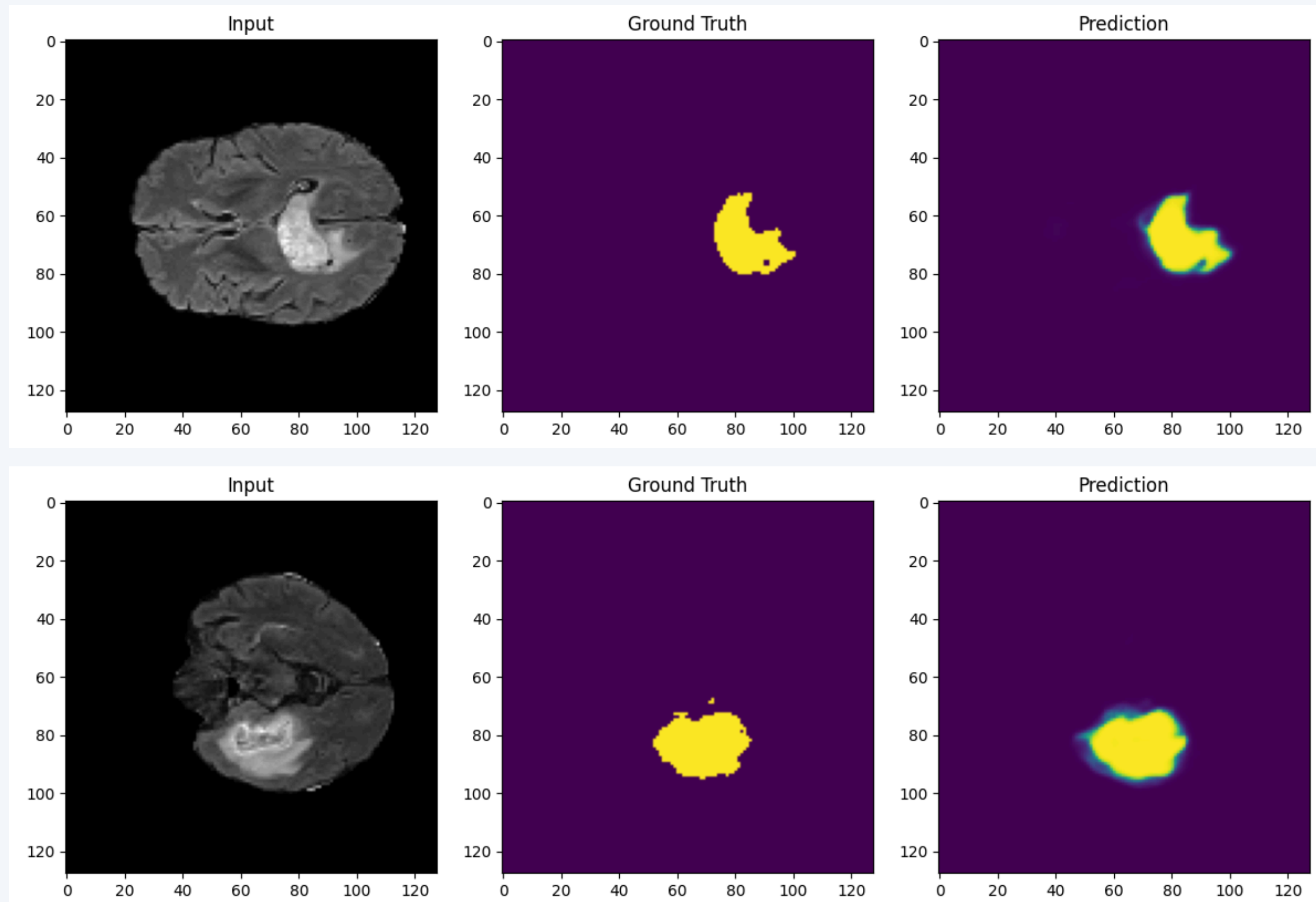
- Better metric of success for segmentation than accuracy
- Measures overlap between two sets

U-Net Training Curve

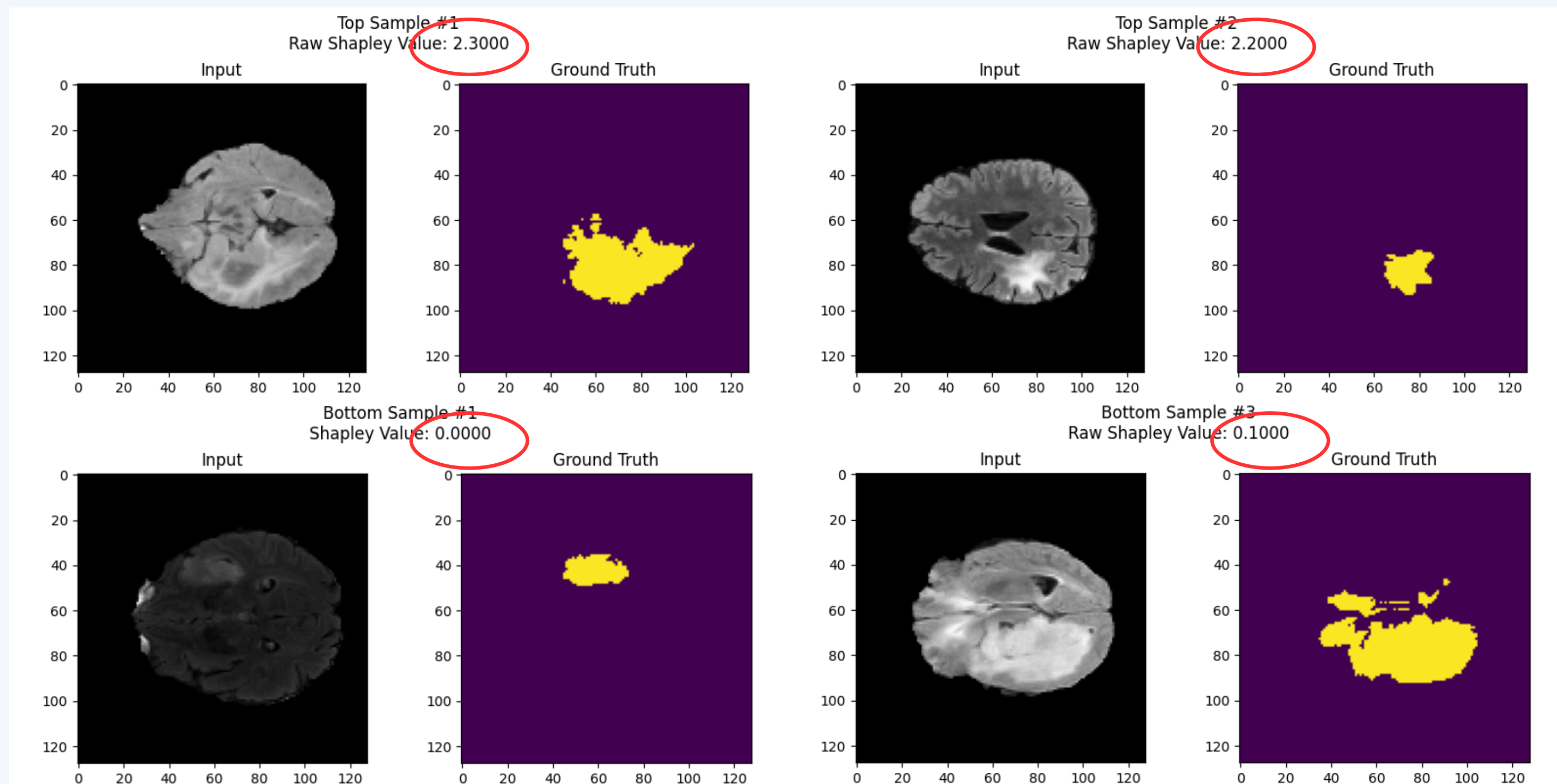


Hyperparameters:
Max Epoch: 100
Batch Size: 32
Learning Rate: 5e-04
Early Stop at ~ 42 epoch

U-Net Model Sample Results

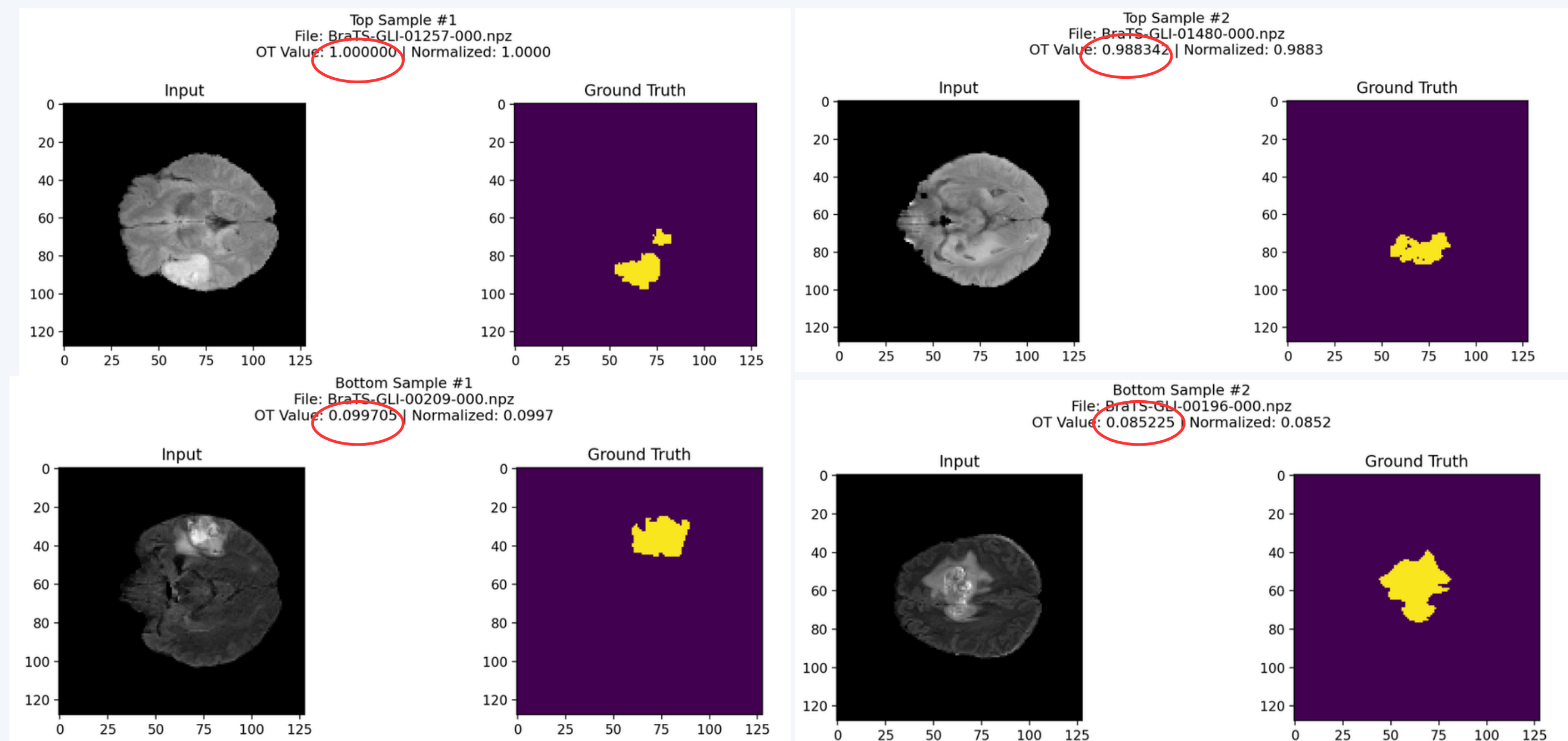


Meets specification: Importance Values are Assigned to Data Points



Shapely Results

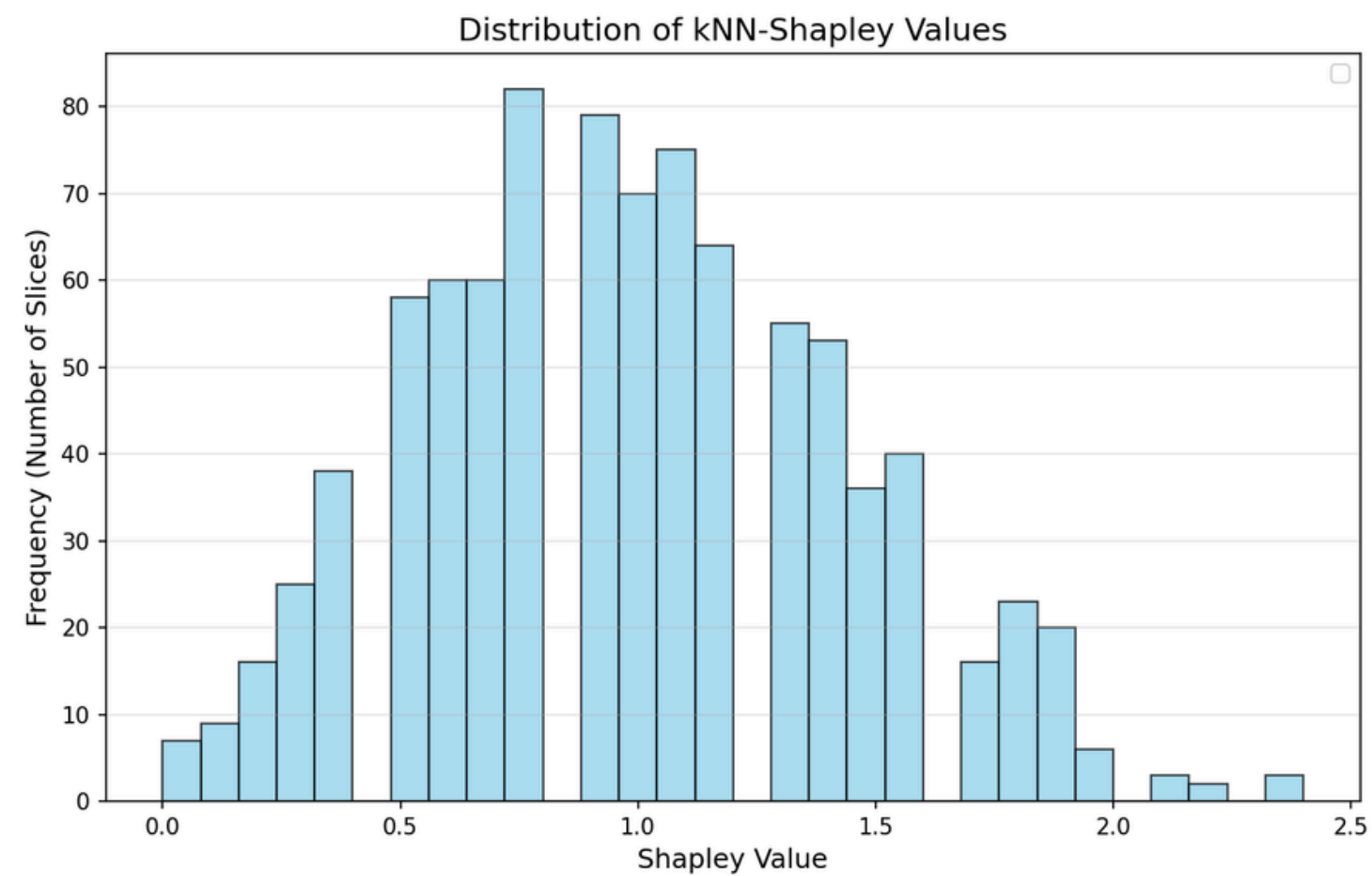
Meets specification: Importance Values are Assigned to Data Points



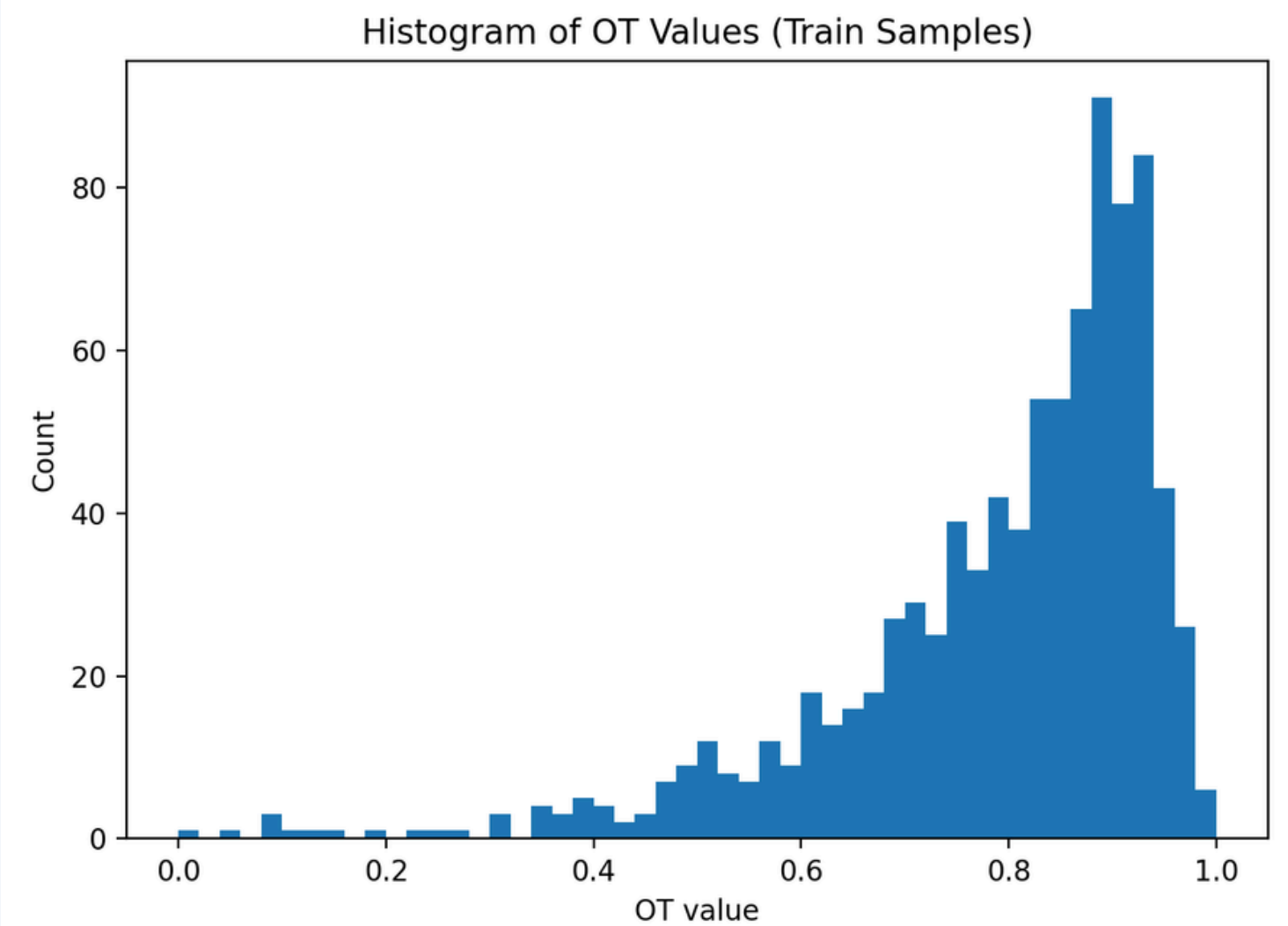
OT Results

Distribution of Values

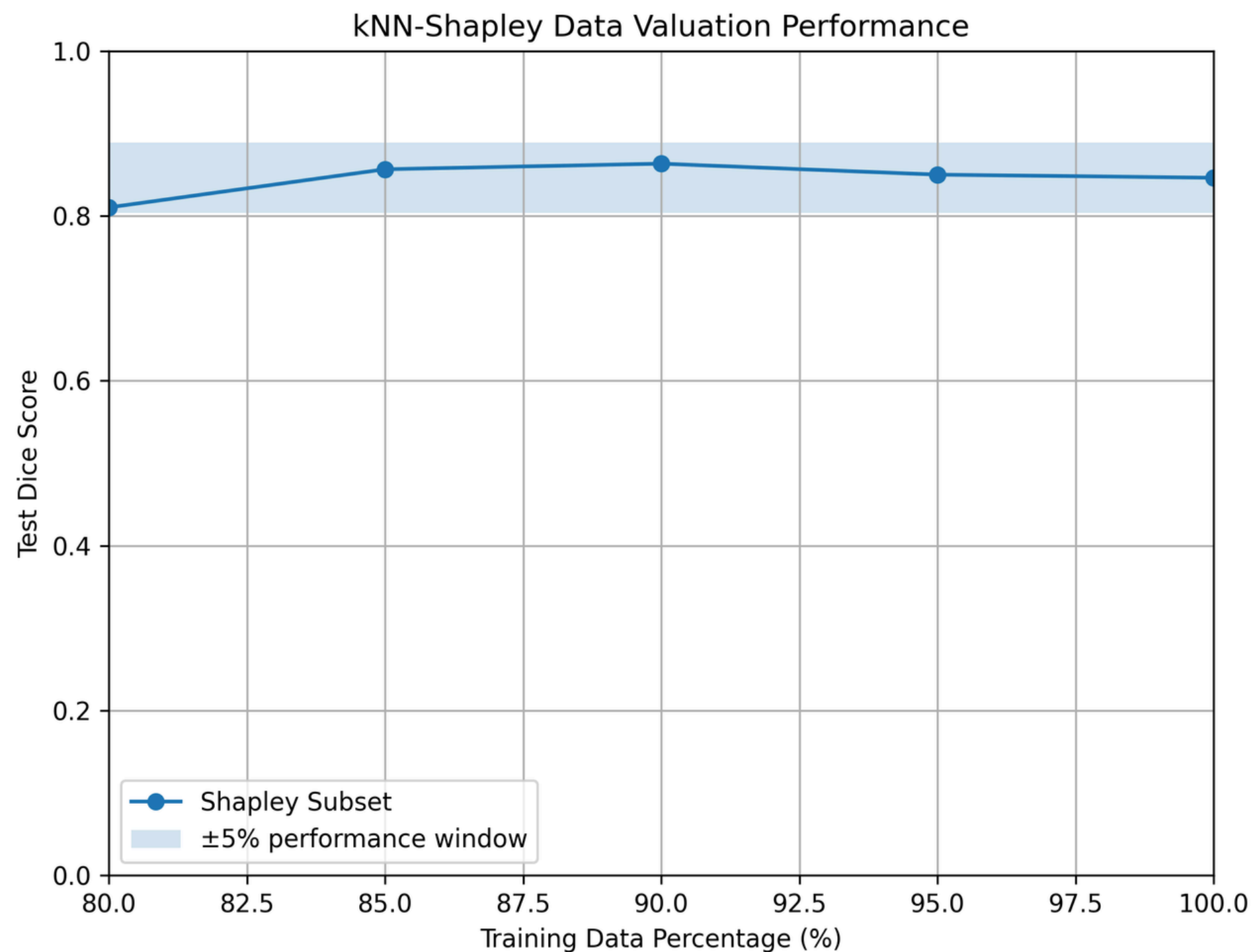
Shapley Values



Optimal Transport



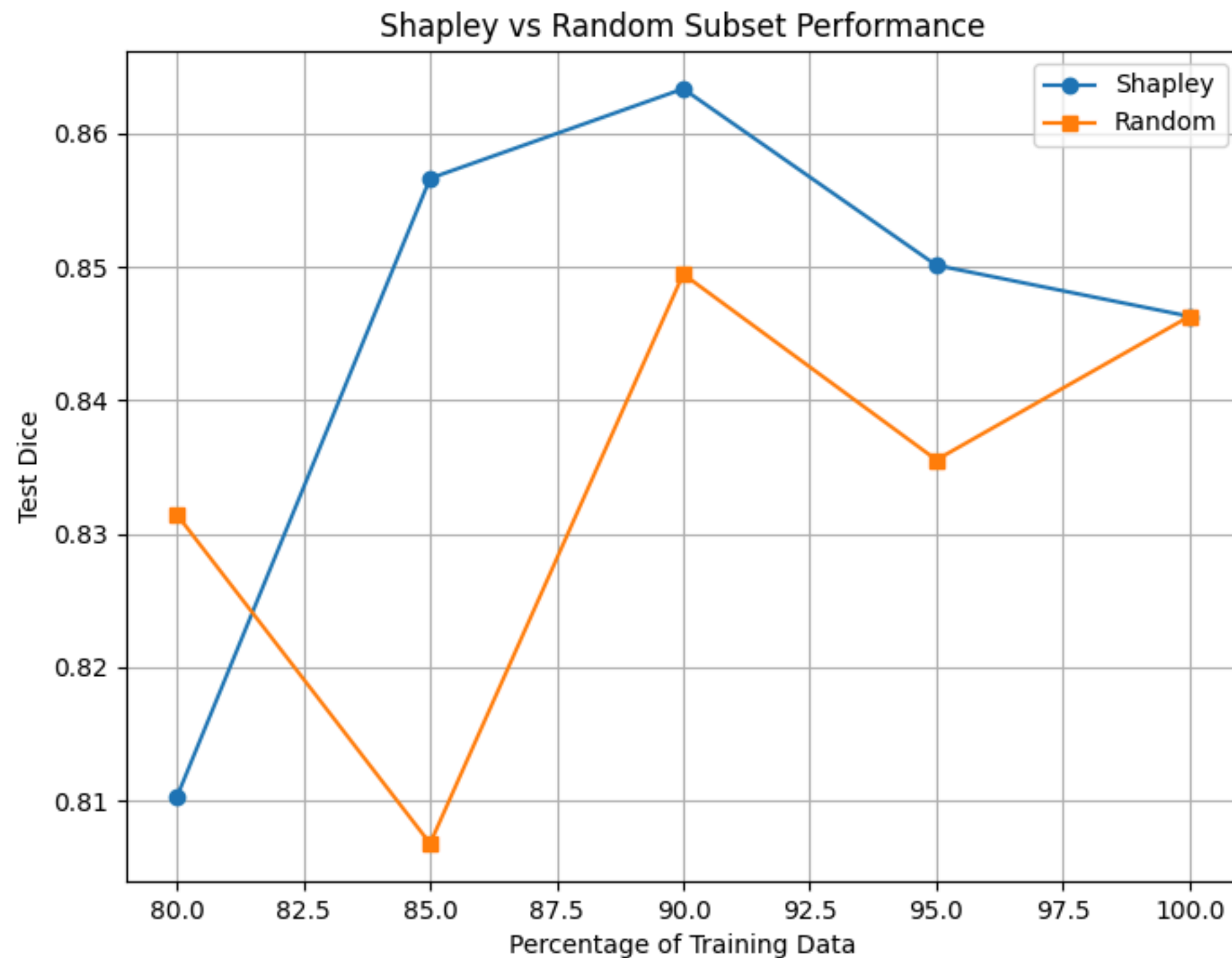
Dice Score vs. Percentage of Top Shapely Data



Meets requirement:

- The system should help decrease use of medical imaging data for ML/DL use
- The system should not reduce the ML model performance by more than 5% while reducing at least 10% of the data size.

Distinguishing Data Utility from Random Chance



Overall Shapely subsets out-performs random pruning

T-test: not significant (low degrees of freedom with only 4 sets of comparison)

04 Scope, Limitations, & Next Steps

Limitations and Strength

Strengths

- Thorough coverage:
 - 2 DV methods
 - 2 Tasks
 - 2 Modalities
- Satisfied major client need:
 - Highly retain model performance with even 20% of data reduction
- Approach to validate findings through random sets

Limitations

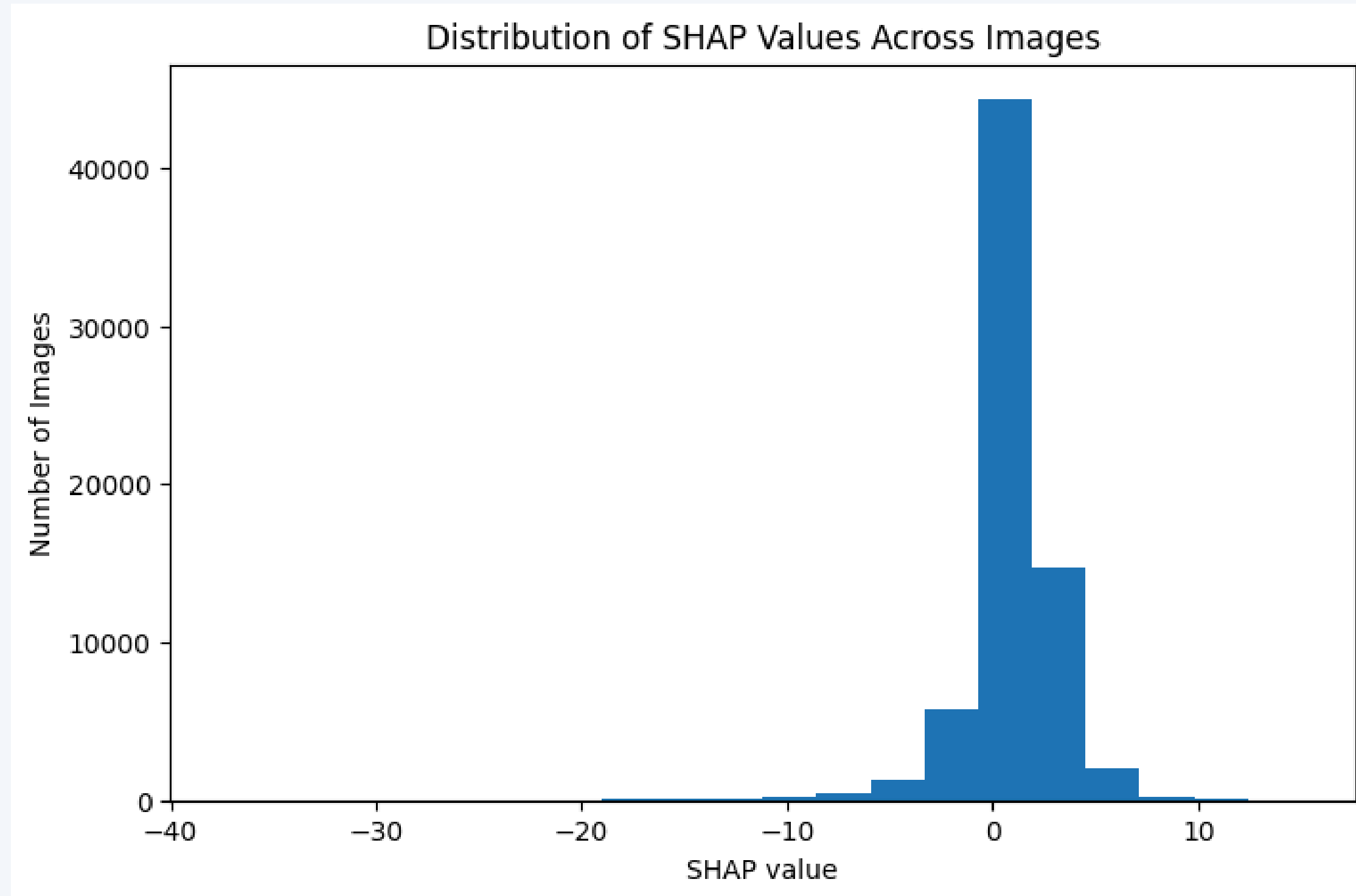
- Underlying mechanism behind results not fully understood
- Require consistency analysis of results

Next Steps

- Additional fine tuning and alignment of shapely and OT pipelines
- Full 4 pipeline integration for open source standard ease of use on GitHub
 - Flags are already included in some of the scripts, which identify:
 - Task
 - DV type
 - Modality
- What data points were removed?
 - Was it noise or was it diversity of images?
- Stretch goal: Evaluate environmental impact using CodeCarbon

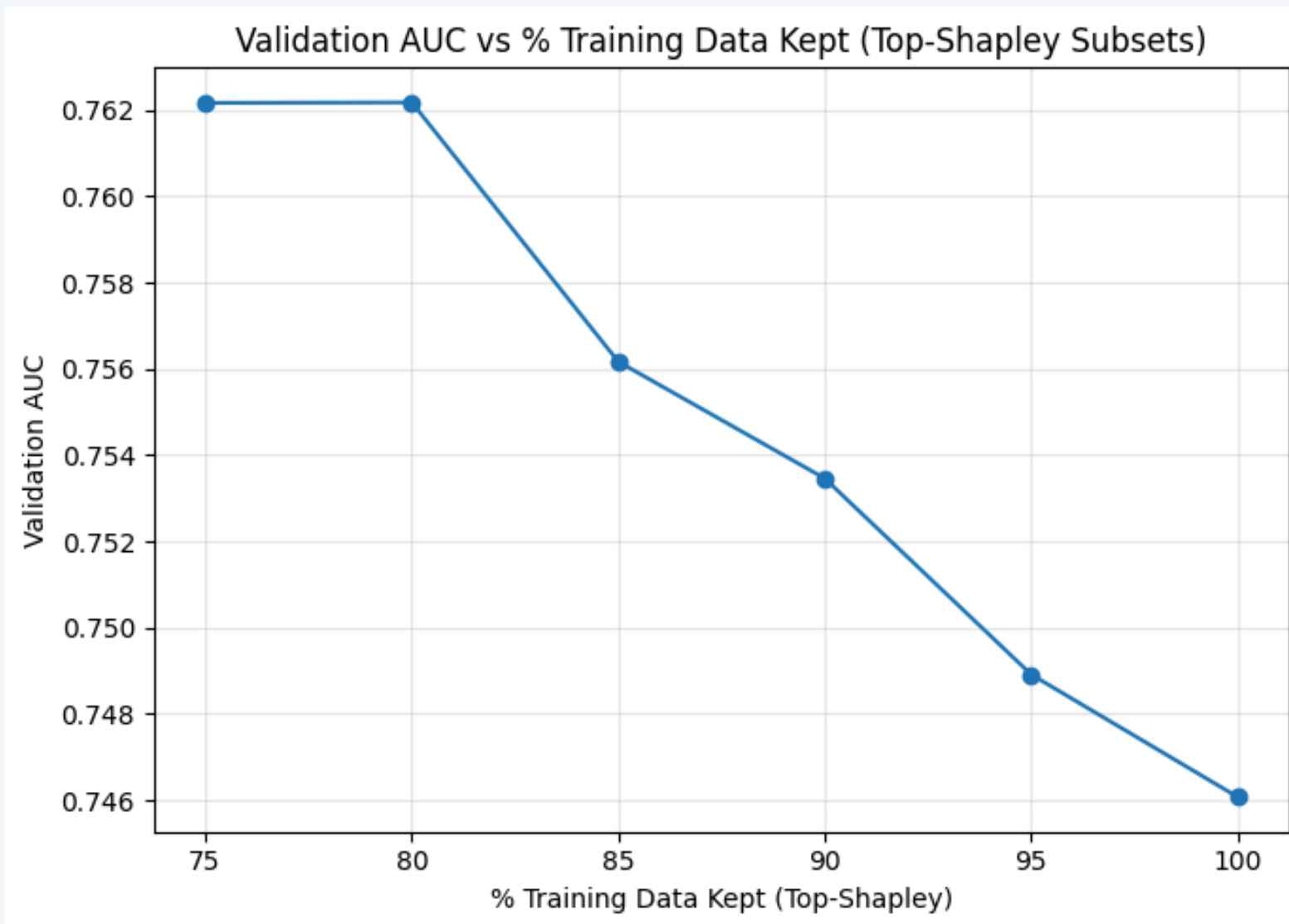
Appendix

Appendix



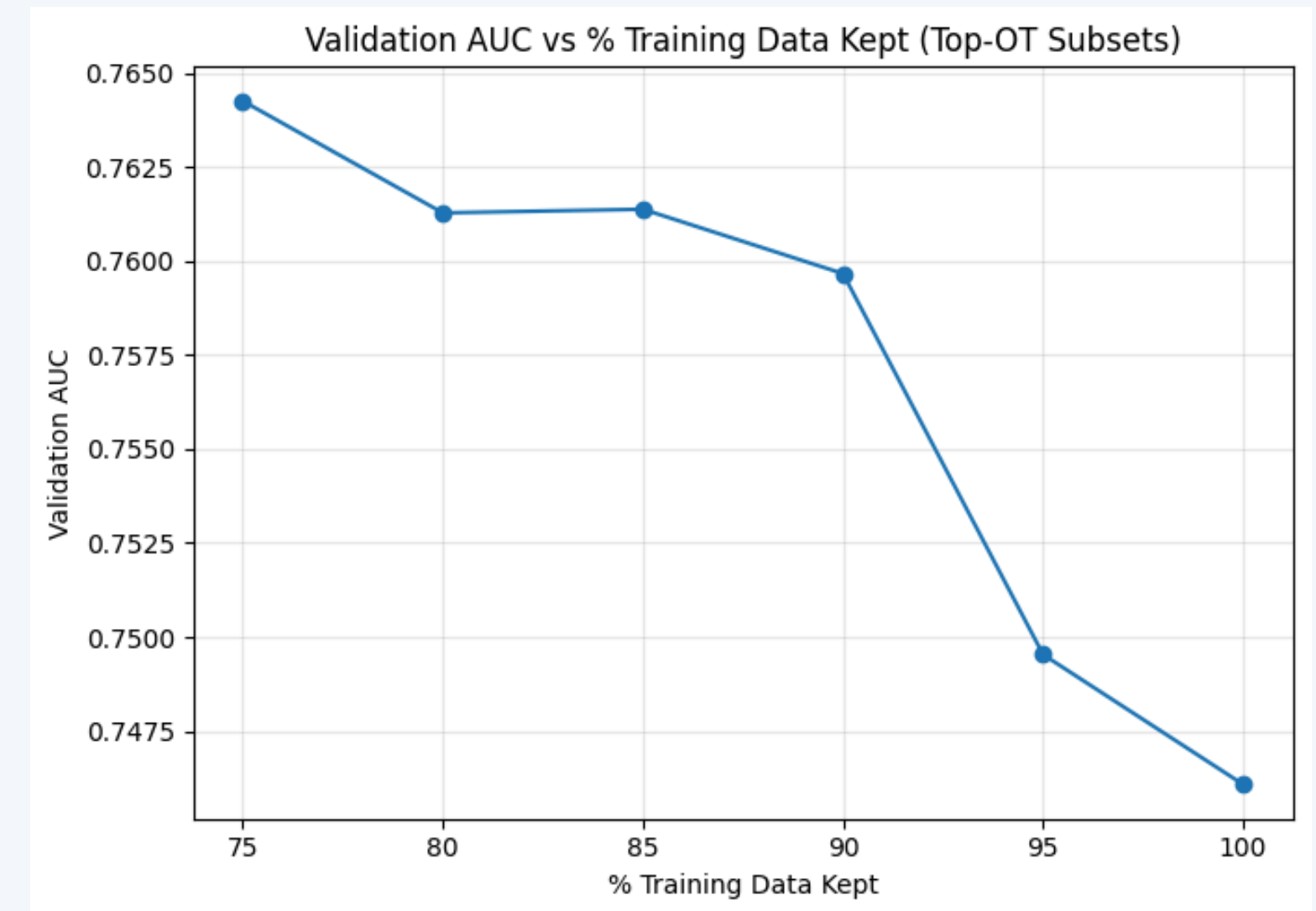
Validation AUC vs % Training Data Kept

Shapley Values

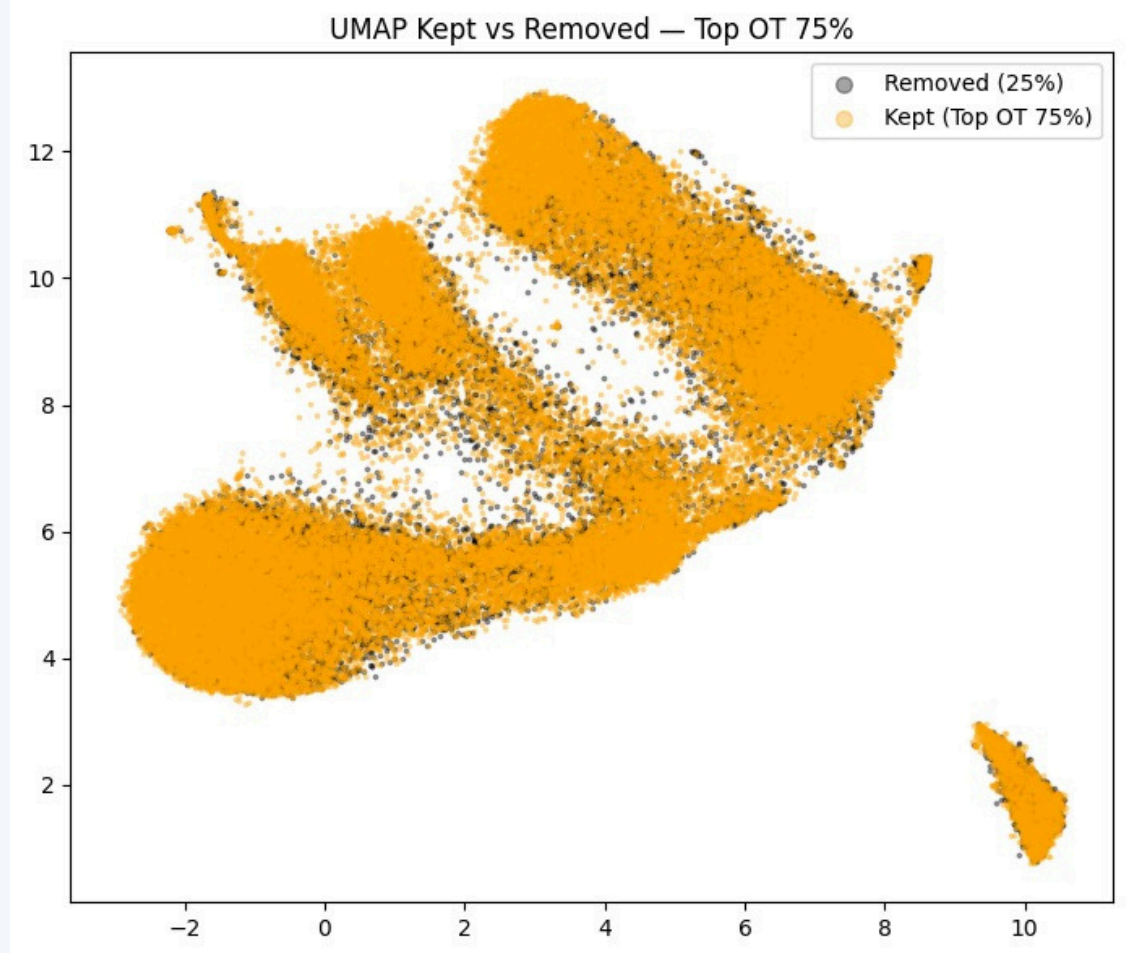
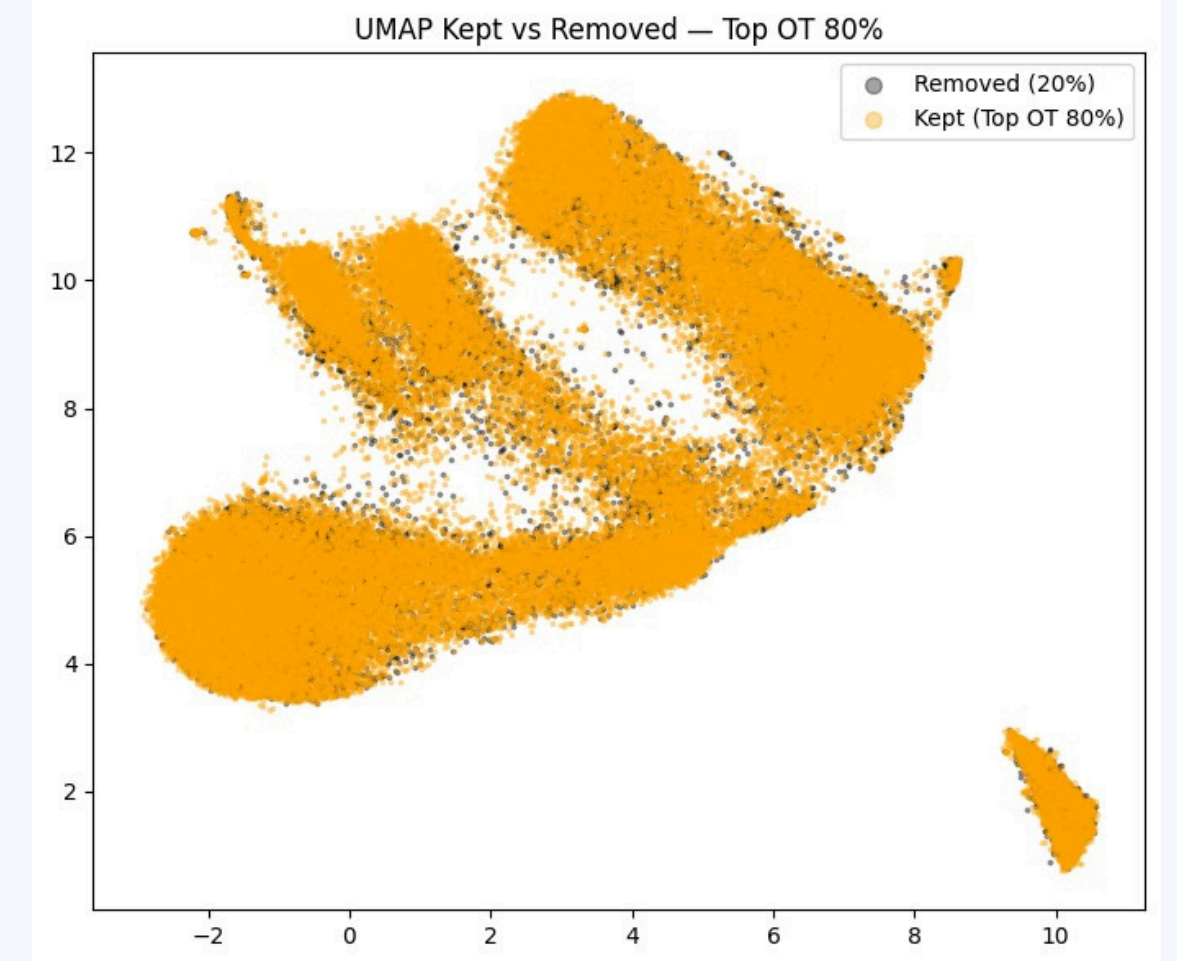
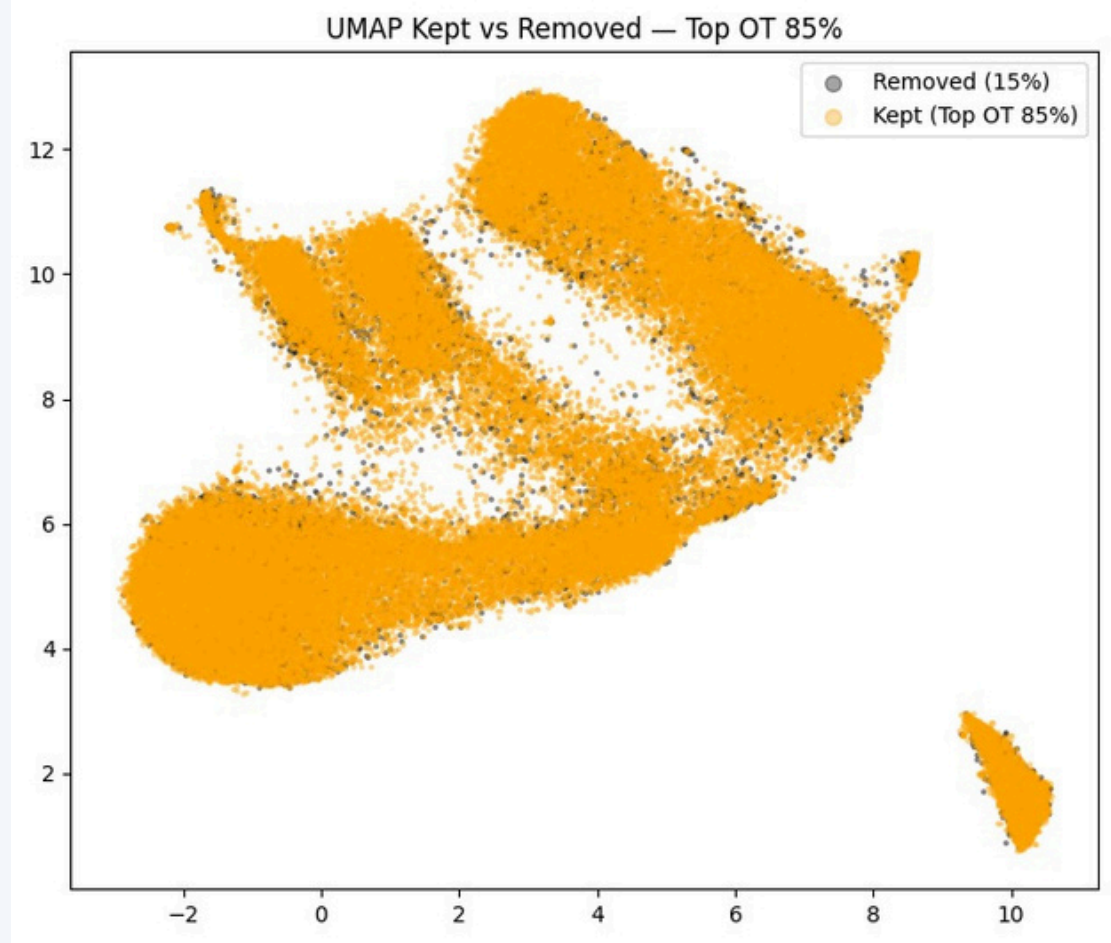
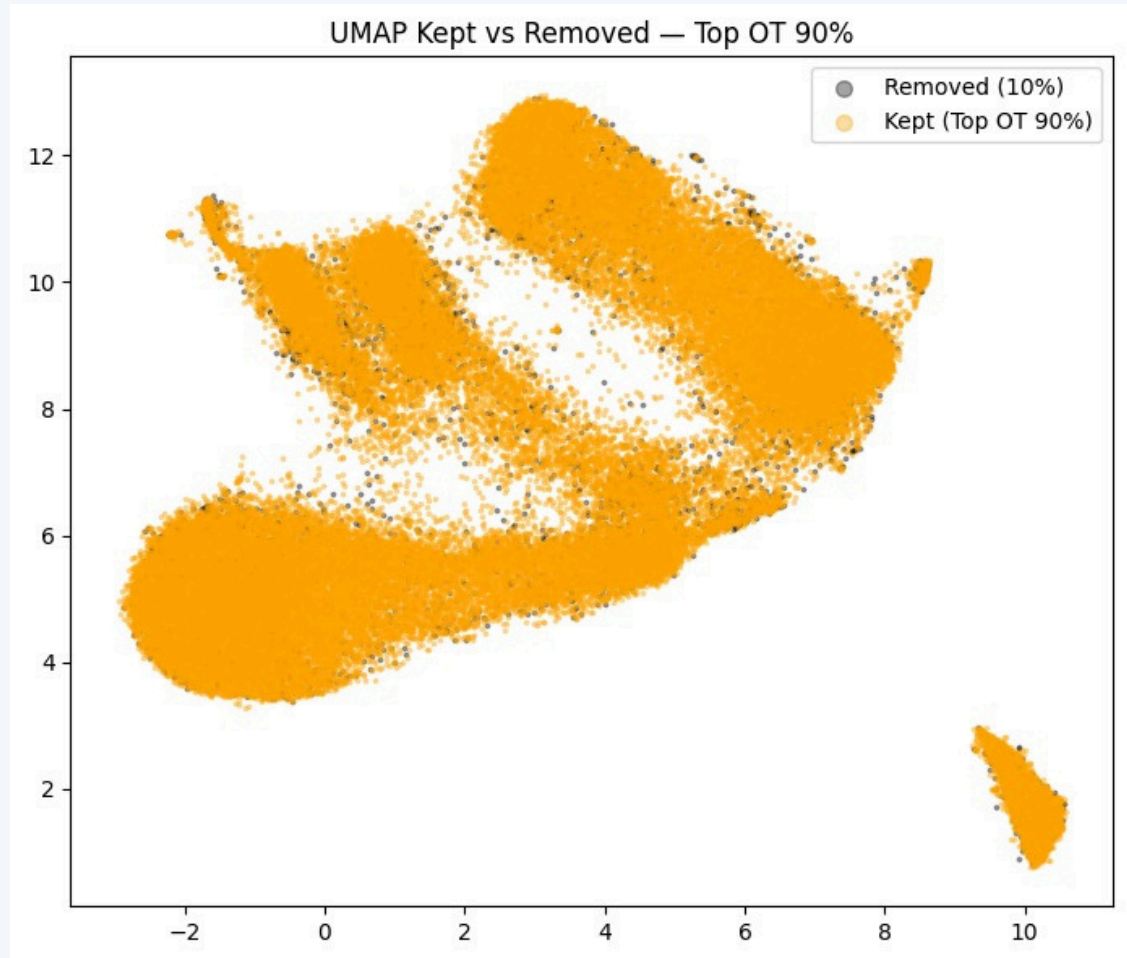
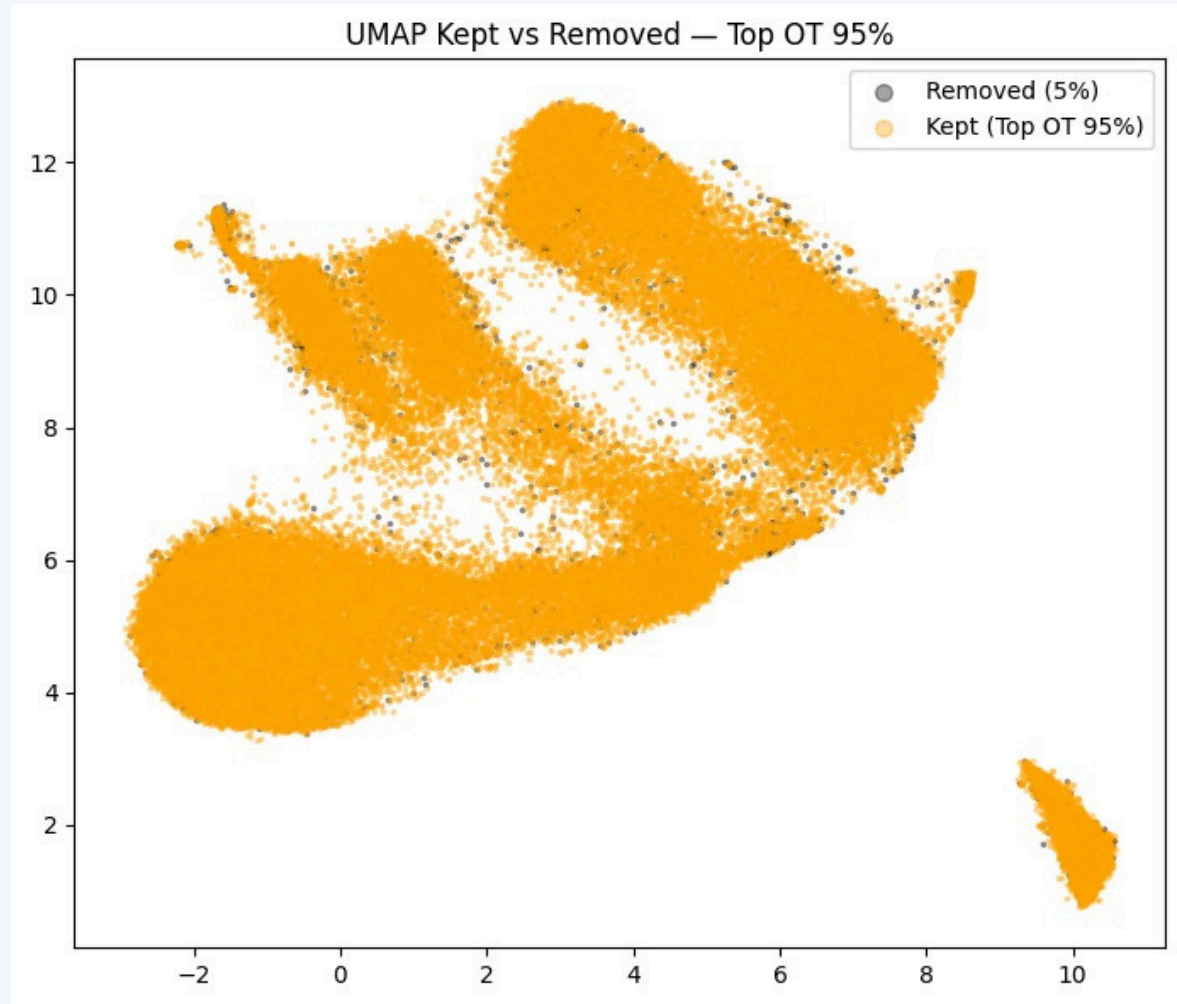
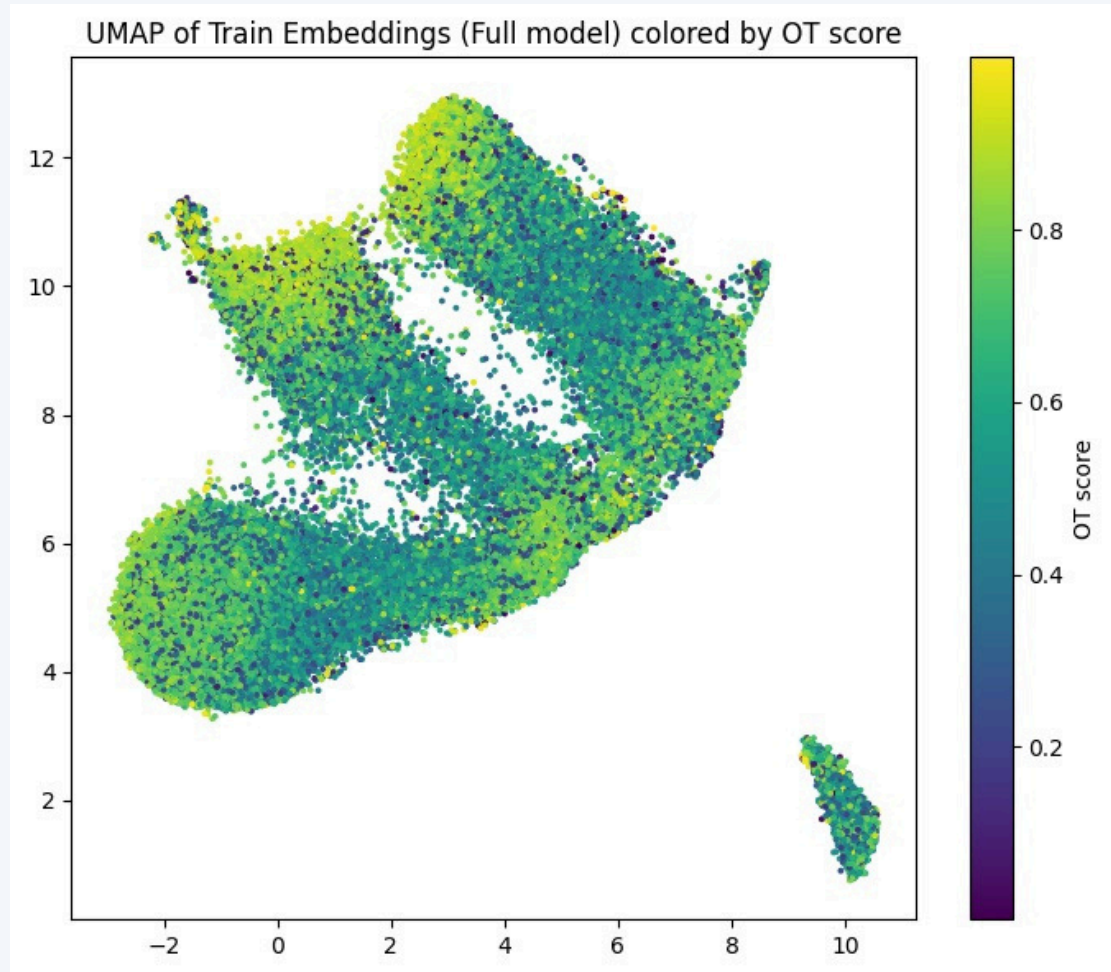


Max val AUC = 0.762179 at 80% data kept
Min val AUC = 0.746078 at 100% data kept
Difference (max - min) = 0.016101

Optimal Transport



Max Val AUC : 0.764256 at 75% data kept
Min Val AUC : 0.746078 at 100% data kept
Difference(max - min) = 0.018178



UMAP of Kept Points — Top OT 95%



UMAP of Kept Points — Top OT 85%



UMAP of Kept Points — Top OT 75%

